



***National University of Computer and
Emerging Sciences (FAST-NUCES)***

PF POST-LAB TASK

LAB-04

C PROGRAMMING

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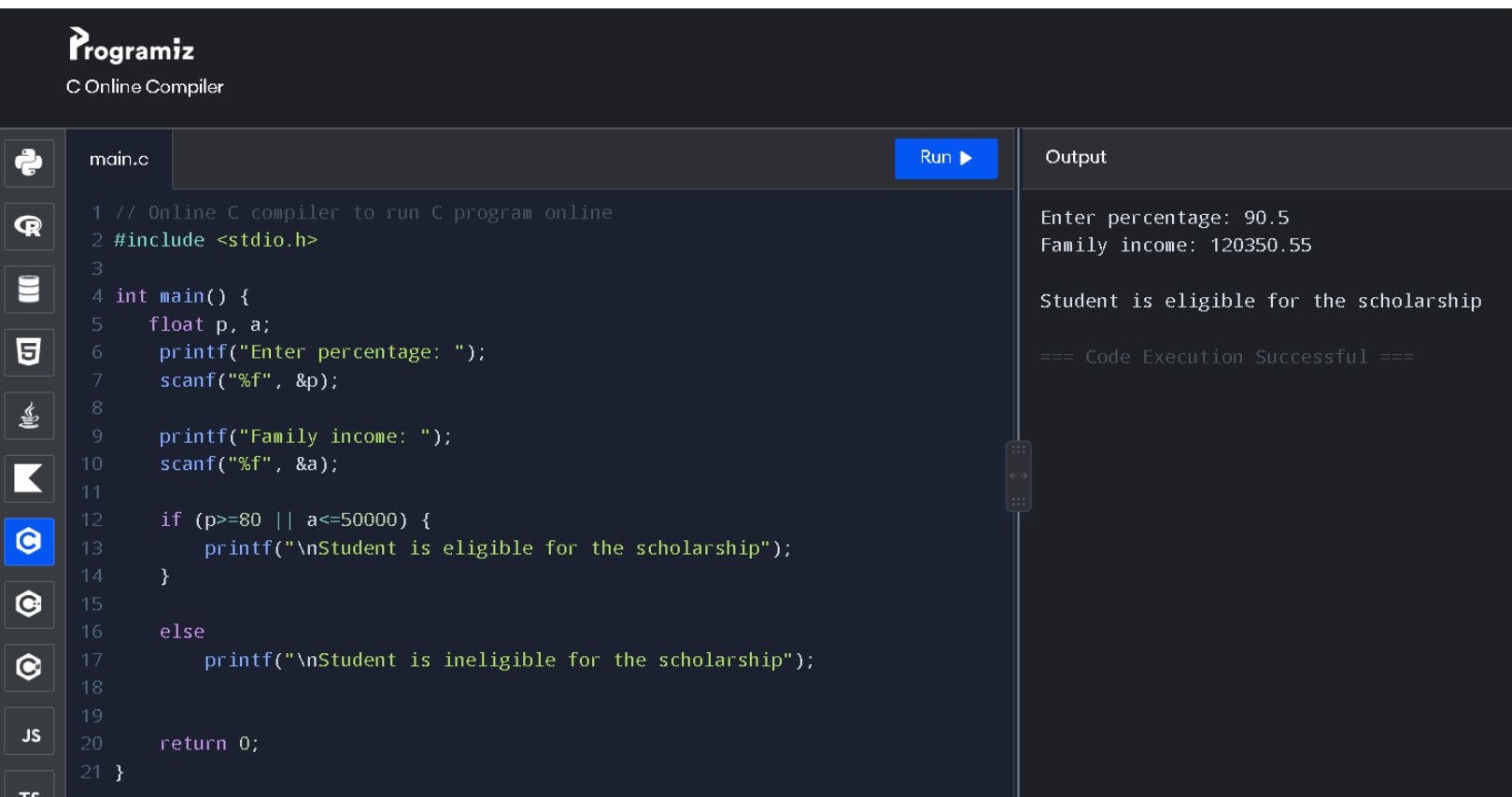
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BSE-1C

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Problem:

A university offers a scholarship to students who have either obtained 80% or above marks OR have a family income below Rs. 50,000. Write a C program that takes marks and family income as input using scanf(). Use relational operators, the OR (||) operator, and if-else to determine whether the student qualifies for the scholarship.



The screenshot shows the Programiz C Online Compiler interface. On the left, there is a sidebar with icons for various programming languages: Python, R, Java, C, JavaScript, and TypeScript. The main editor area displays a C program in a file named 'main.c'. The code is as follows:

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     float p, a;
6     printf("Enter percentage: ");
7     scanf("%f", &p);
8
9     printf("Family income: ");
10    scanf("%f", &a);
11
12    if (p>=80 || a<=50000) {
13        printf("\nStudent is eligible for the scholarship");
14    }
15
16    else
17        printf("\nStudent is ineligible for the scholarship");
18
19
20    return 0;
21 }
```

On the right side of the editor, there is a 'Run' button and an 'Output' panel. The output panel shows the following text:

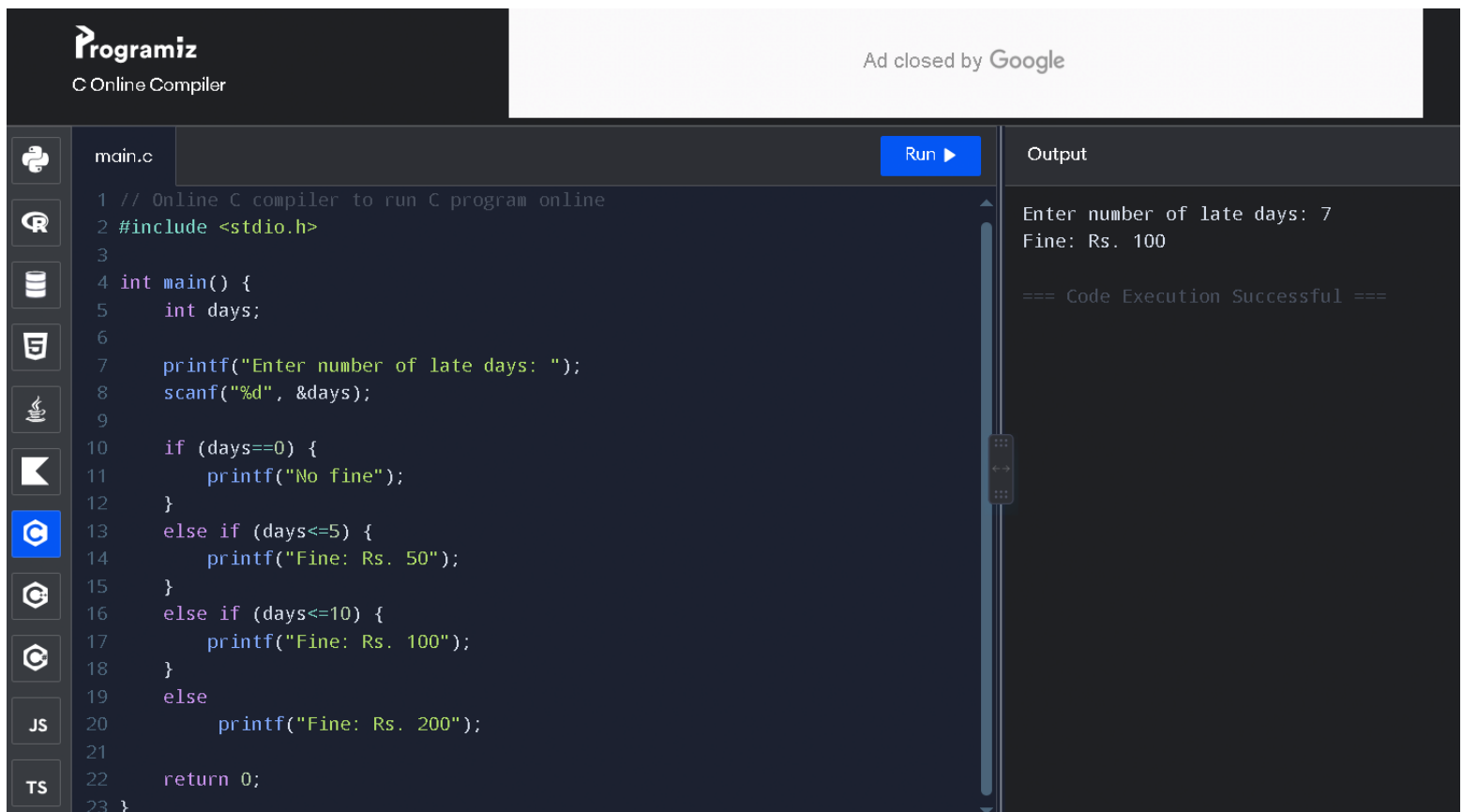
```
Enter percentage: 90.5
Family income: 120350.55

Student is eligible for the scholarship

=== Code Execution Successful ===
```

Problem:

A library wants to calculate a student's fine based on the number of late days. Ask the user to enter the number of late days using `scanf()`. If the student has 0 days, display "No Fine". If the late days are 1–5, display "Fine: Rs. 50". If the late days are 6–10, display "Fine: Rs. 100". For more than 10 days, display "Fine: Rs. 200". Use relational operators and nested if-else conditions.



The screenshot shows the Programiz C Online Compiler interface. The top bar includes the Programiz logo and the text "C Online Compiler". An advertisement for Google is visible on the right. The main editor area displays a C program named `main.c` with the following code:

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int days;
6
7     printf("Enter number of late days: ");
8     scanf("%d", &days);
9
10    if (days==0) {
11        printf("No fine");
12    }
13    else if (days<=5) {
14        printf("Fine: Rs. 50");
15    }
16    else if (days<=10) {
17        printf("Fine: Rs. 100");
18    }
19    else
20        printf("Fine: Rs. 200");
21
22    return 0;
23 }
```

A blue "Run" button is located to the right of the code editor. The output panel on the right shows the execution results:

```
Enter number of late days: 7
Fine: Rs. 100

=== Code Execution Successful ===
```

Problem:

A student registration system needs to collect a student's name. Write a C program that uses `fgets()` to take the student's full name, including spaces. Then use `puts()` to display a message containing the entered name. The program should demonstrate the difference between taking a complete line of text and single character.

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main.c

Run ▶

```
1 #include <stdio.h>
2
3 int main() {
4     char name[40];
5     printf("Use of fgets:-");
6     printf("\nEnter student full name: ");
7     fgets(name, sizeof(name), stdin);
8     //fgets function will read complete line of text
9
10    puts("student reistered name is:");
11    puts(name);
12    printf("\nLimitation of scanf:-\n");
13    printf("Enter yuor full name: ");
14    scanf("%s", name);
15    //scanf function will read a single character of complete line of text
16
17    printf("student reistered name is: ");
18    printf("%s", name);
19
20    return 0;
21 }
```

Output

Use of fgets:-
Enter student full name: Syed Zain Ali
student reistered name is:
Syed Zain Ali

Limitation of scanf:-
Enter yuor full name: Syed Zain Ali
student reistered name is: Syed

=== Code Execution Successful ===

Problem:

Write a C program that takes the length and width of a rectangle from the user and calculates its area and perimeter


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main.c

Run ▶

```
1 #include <stdio.h>
2
3 int main() {
4     float l,w,A,P;
5     printf("Length of rectangle= ");
6     scanf("%f", &l);
7
8     printf("Width of rectangle= ");
9     scanf("%f", &w);
10
11     A=l*w;
12     printf("\nArea of rectangle= %.2f square units\n", A);
13     P=2*l+2*w;
14     printf("Perimeter of rectangle= %.2f units", P);
15
16     return 0;
17 }
```

Output

Length of rectangle= 7
Width of rectangle= 7.5

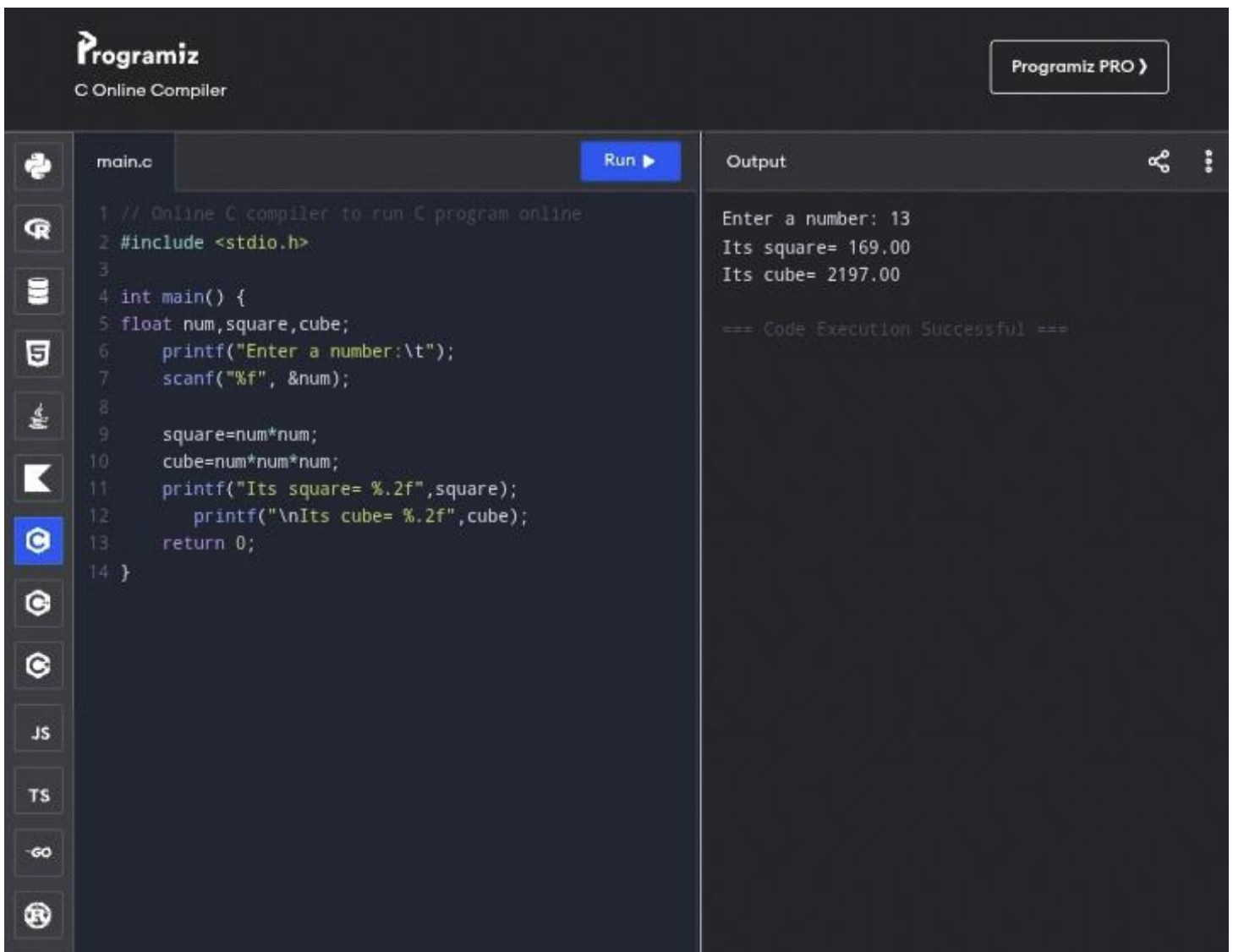
Area of rectangle= 52.50 square units
Perimeter of rectangle= 29.00 units

=== Code Execution Successful ===

JS

Problem:

Write a C program that takes an integer from the user and calculates and displays its square and cube



The screenshot shows the Programiz C Online Compiler interface. The top left features the Programiz logo and the text "C Online Compiler". The top right has a "Programiz PRO" button. The main area is divided into two panels: a code editor on the left and an output panel on the right. The code editor shows a C program named "main.c" with the following code:

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     float num,square,cube;
6     printf("Enter a number:\t");
7     scanf("%f", &num);
8
9     square=num*num;
10    cube=num*num*num;
11    printf("Its square= %.2f",square);
12    printf("\nIts cube= %.2f",cube);
13    return 0;
14 }
```

The output panel shows the result of running the program:

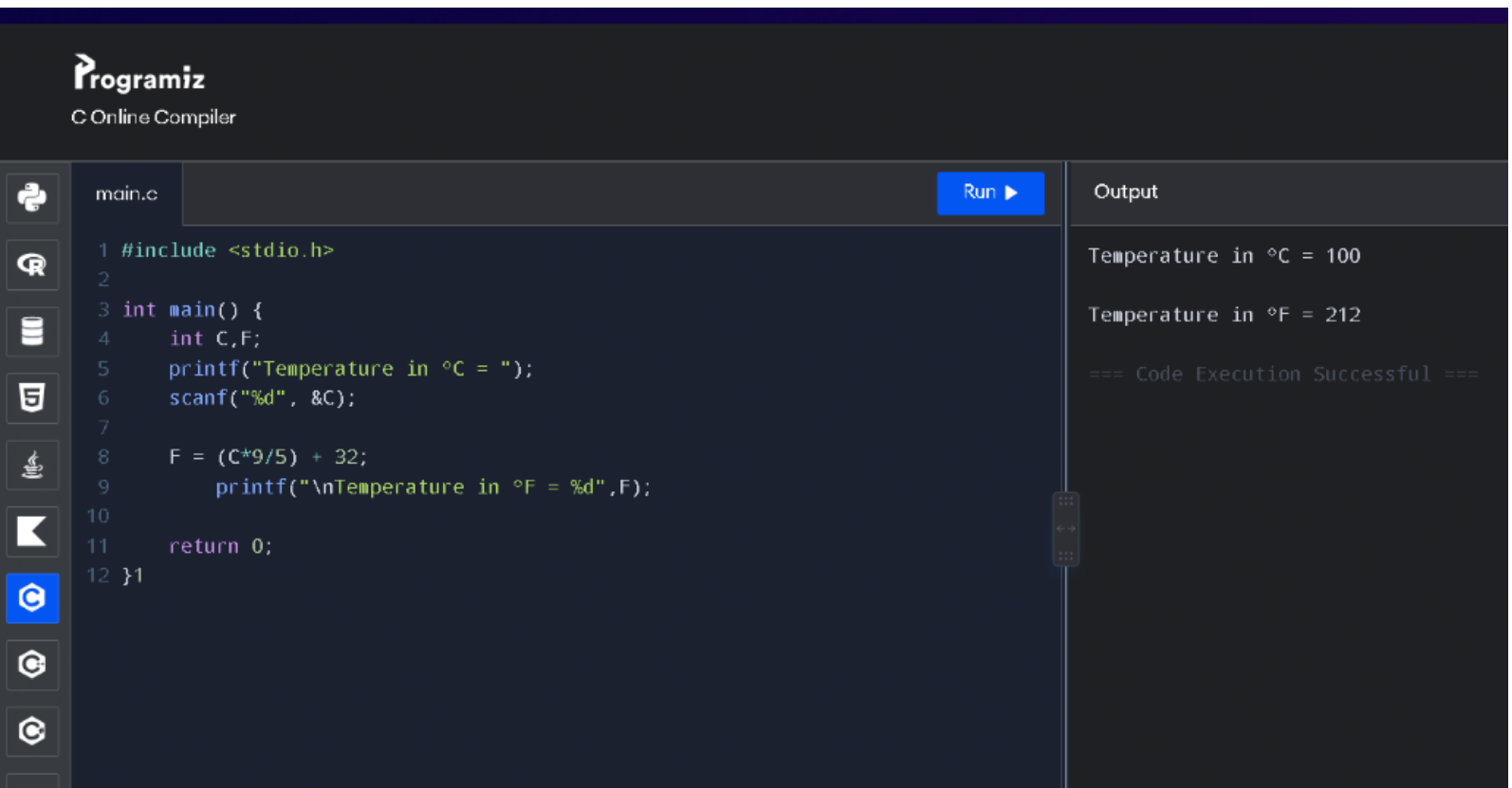
```
Enter a number: 13
Its square= 169.00
Its cube= 2197.00

=== Code Execution Successful ===
```

On the left side of the code editor, there is a vertical toolbar with icons for various programming languages and a "GO" button. The languages listed are JS, TS, and GO.

Problem:

Write a C program that takes temperature in Celsius from the user and converts it into Fahrenheit using the formula: $F = (C \times 9/5) + 32$



The screenshot displays the Programiz C Online Compiler interface. On the left, a vertical toolbar contains icons for various programming languages: Python, R, Java, JavaScript, C, C++, C#, PHP, and Ruby. The main editor area shows a C program in a file named 'main.c'. The code includes the standard input/output header, declares variables for Celsius and Fahrenheit, prompts the user for input, and performs the conversion using the formula $F = (C \times 9/5) + 32$. A 'Run' button is located at the top right of the editor. To the right of the editor, the 'Output' panel shows the program's execution results, including the prompts and the calculated Fahrenheit values for an input of 100. The output concludes with a success message.

```
1 #include <stdio.h>
2
3 int main() {
4     int C,F;
5     printf("Temperature in °C = ");
6     scanf("%d", &C);
7
8     F = (C*9/5) + 32;
9     printf("\nTemperature in °F = %d",F);
10
11     return 0;
12 }
```

Output

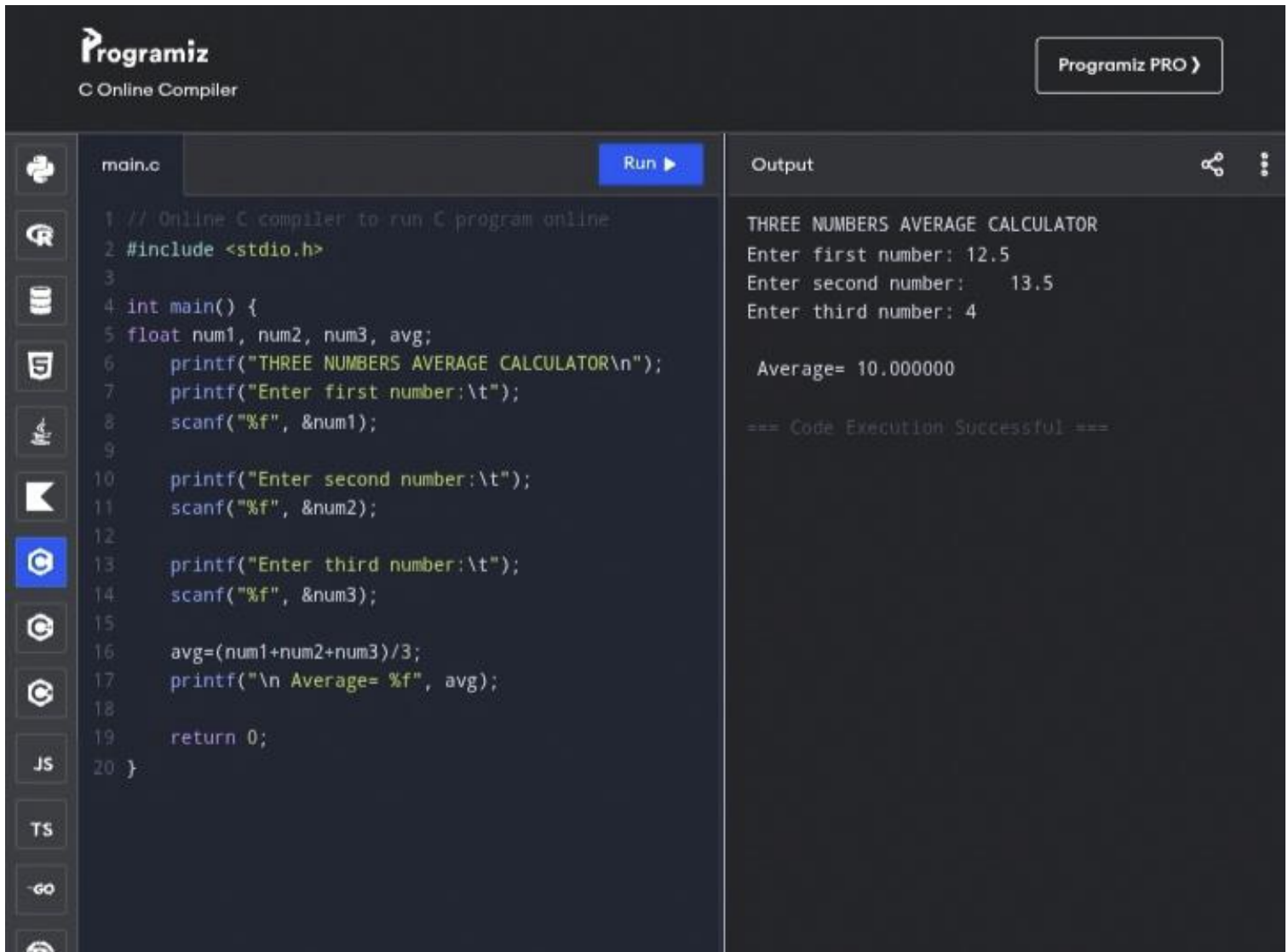
Temperature in °C = 100

Temperature in °F = 212

=== Code Execution Successful ===

Problem:

Write a C program that takes three numbers from the user and calculates and displays their average.



The screenshot shows the Programiz C Online Compiler interface. The left sidebar contains icons for various programming languages, with C selected. The main editor displays a C program named 'main.c' with the following code:

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     float num1, num2, num3, avg;
6     printf("THREE NUMBERS AVERAGE CALCULATOR\n");
7     printf("Enter first number:\t");
8     scanf("%f", &num1);
9
10    printf("Enter second number:\t");
11    scanf("%f", &num2);
12
13    printf("Enter third number:\t");
14    scanf("%f", &num3);
15
16    avg=(num1+num2+num3)/3;
17    printf("\n Average= %f", avg);
18
19    return 0;
20 }
```

A blue 'Run' button is located to the right of the code editor. The right sidebar shows the 'Output' window, which displays the program's execution results:

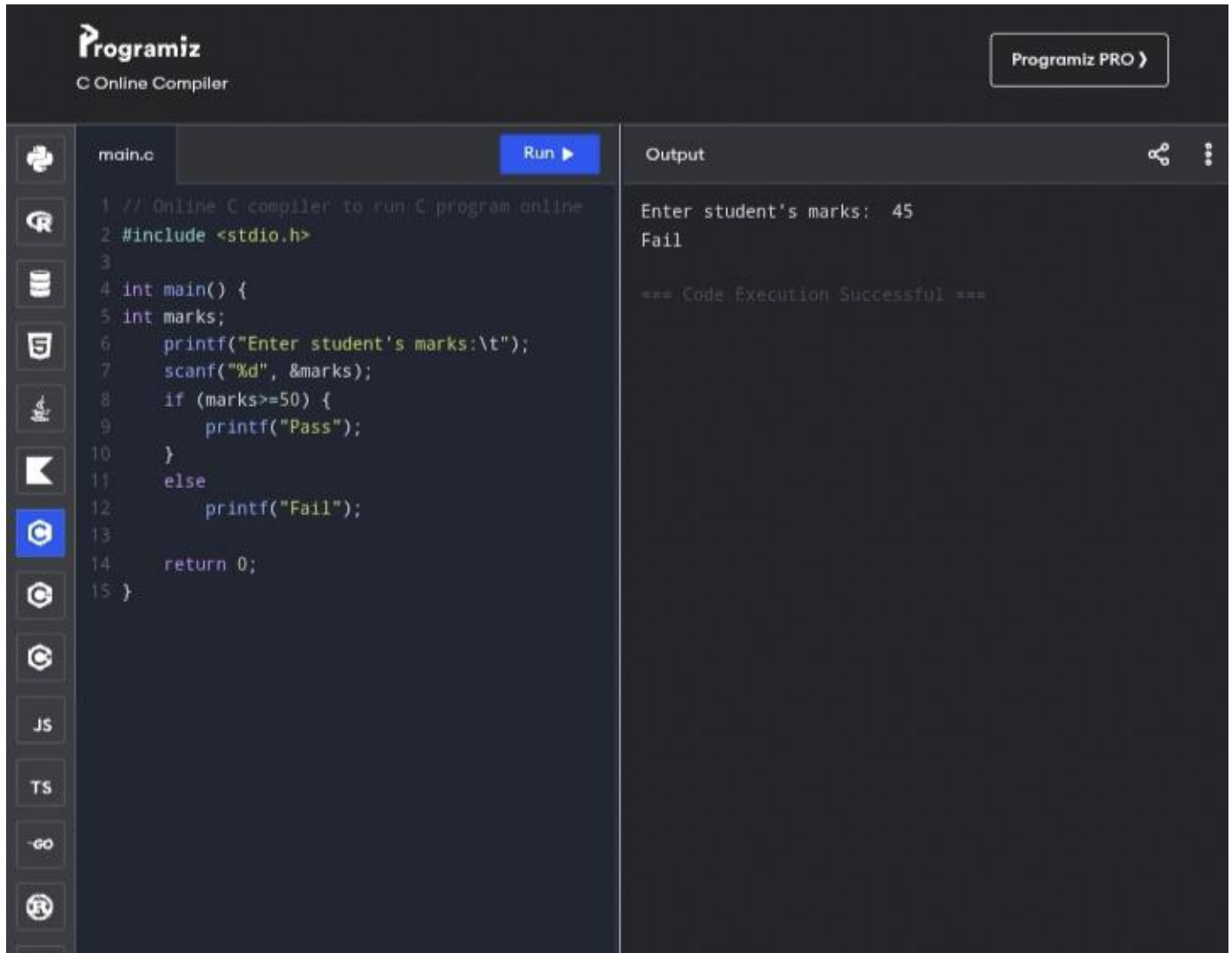
```
THREE NUMBERS AVERAGE CALCULATOR
Enter first number: 12.5
Enter second number: 13.5
Enter third number: 4

Average= 10.000000

=== Code Execution Successful ===
```


Problem:

Write a C program that takes a student's marks and displays "Pass" if the marks are 50 or above; otherwise, display "Fail".



The screenshot shows the Programiz C Online Compiler interface. The left pane displays the C code for a program named `main.c`. The code prompts the user to enter student marks, reads the input, and checks if it is 50 or above to determine the result. The right pane shows the output of the program, which is "Fail" for an input of 45, followed by a success message.

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int marks;
6     printf("Enter student's marks:\n");
7     scanf("%d", &marks);
8     if (marks >= 50) {
9         printf("Pass");
10    }
11    else
12        printf("Fail");
13
14    return 0;
15 }
```

Output

Enter student's marks: 45
Fail

=== Code Execution Successful ===